

DBDOME — Step-by-Step Installation & Configuration Guide

Applies to: DBDOME installer v2.0.0 (`dbdome_setup.exe`) on Windows Server 2022 / 2025 (x64)
Covers: mounting the media, installation (console + PowerShell), first login, adding a monitored server, whitelisting an applicative login, mail configuration, and updating.

1. Before You Start

1.1 Server requirements

Item	Requirement
OS	Windows Server 2022 or 2025, x64
CPU / RAM / Disk	4 cores / 16 GB / 250 GB (SSD or NVMe recommended)
Privileges	Local Administrator (the installer refuses to run otherwise)
Free TCP ports	8080 (DBDOME Web UI), 3000 (Grafana), 5432 (PostgreSQL)
Network	Outbound reachability to every monitored database on its listener port (SQL Server 1433, Oracle 1521, PostgreSQL 5432, MySQL/MariaDB 3306, ClickHouse 8123)
Static IP	Strongly recommended — the installer records the machine's LAN IP as <code>ORG_IP</code> and uses it in dashboard links

Nothing else needs pre-installing: Python is embedded in the installer, PostgreSQL 18 and the Microsoft ODBC Driver for SQL Server are bundled and installed automatically. **Oracle monitoring only:** install Oracle Instant Client yourself and point `ORACLE_CLIENT_LIB_DIR` in `C:\ProgramData\DBDOME\bin\.env` at it (e.g. `C:\oracle\instantclient_23_0`).

1.2 What gets installed

Component	Detail
Services	<code>DBDOME_scheduler</code> + <code>DBDOME_web</code> (older builds: one combined <code>DBDOME_dbanalytics</code>), <code>DBDOME_Grafana</code> (NSSM-wrapped), <code>postgresql-x64-18</code> — all Automatic start with crash-restart
Install root	<code>C:\ProgramData\DBDOME</code> (binaries in <code>bin\</code> , Grafana config in <code>conf\</code> , <code>grafana.db</code> in <code>data\</code>)
PostgreSQL	<code>C:\Program Files\PostgreSQL\18</code> , port 5432 , database <code>dbanalytics</code>

Component	Detail
DB roles	postgres , dbdome_mon_usr , dbdome_adm — initial password Yd2243796Anz!! (change after install, see §7.1)
Firewall	Inbound rules "DBDOME HTTP Server" (8080), "DBDOME Grafana" (3000), "DBDOME PostgreSQL" (5432) for Domain+Private profiles
Shortcut	DBDOME.lnk on the Public desktop → the Web UI

2. Mount the Installation Media

The installer looks for a payload folder named `DBDOME` (it must contain a `bin\` subfolder). It searches, in order: next to `dbdome_setup.exe` → a DVD volume labeled `DBDOME_DVD_ROM` → `<any drive>:\DBDOME`.

Option A — ISO file (PowerShell, run as Administrator):

```
Mount-DiskImage -ImagePath "C:\path\to\DBDOME.iso"
# note the drive letter it was assigned:
(Get-DiskImage -ImagePath "C:\path\to\DBDOME.iso" | Get-Volume).DriveLetter
```

Option B — physical DVD: insert the disc (volume label `DBDOME_DVD_ROM`); no mounting needed.

Option C — copied folder: place the `DBDOME` folder in the same directory as `dbdome_setup.exe`, or at the root of any drive (`D:\DBDOME`).

3. Install DBDOME

The installer is a **console program** — there is no wizard and there are no questions. It auto-detects everything and runs 16 steps unattended (copy files → ODBC driver → PostgreSQL → database restore → Grafana → `.env` → firewall → services → validation → shortcut).

3.1 Run it (UI way)

1. Log on as a local Administrator.
2. Right-click `dbdome_setup.exe` → **Run as administrator**.
3. Watch the console; installation ends with a summary banner listing the Web UI, Grafana, and PostgreSQL addresses.

3.2 Run it (PowerShell way)

```
# elevated PowerShell
Start-Process -FilePath "D:\dbdome_setup.exe" -Verb RunAs -Wait
Get-Content C:\ProgramData\DBDOME\install.log -Tail 40
```

3.3 Verify the installation

The installer runs its own validation (PostgreSQL connectivity, root-cause catalog loaded, services RUNNING, `.env` present). To re-check manually:

```
Get-Service DBDOME_scheduler, DBDOME_web, DBDOME_Grafana, postgresql-x64-18
Get-NetTCPConnection -LocalPort 8080,3000,5432 -State Listen | Select LocalPort -Unique
```

All services should be **Running**. Full log: `C:\ProgramData\DBDOME\install.log`.

4. First Login

Interface	URL	Notes
DBDOME Web UI	<code>https://<server-ip>:8080/dbdome</code>	HTTPS with a self-signed certificate — the browser will warn once; proceed. To install your own certificate use the <code>/ssl_certificate</code> page, then restart <code>DBDOME_web</code> .
Grafana dashboards	<code>http://<server-ip>:3000</code>	First login <code>admin / admin</code> (Grafana prompts you to change it). See <code>docs\DBDOME_Grafana_User_Access_Guide.pdf</code> for role setup.

The configuration pages in the next sections are normally reached through buttons on the Grafana dashboards, but can also be opened directly at the URLs given.

5. Configuration

5.1 Add a Monitored Server

1. Open `https://<server-ip>:8080/server-form` (or the "Add Server" button on the Grafana 7 – *Automation and Workflows* dashboard).
2. Fill in the form:

Field	Value
Action	Add (the same form does Update / Delete)
IP Address	address of the database host
Server	display name for the server
Port	listener port (1433 / 1521 / 5432 / 3306 / 8123)
DB Vendor	<code>mssql</code> , <code>Oracle</code> , <code>postgres</code> , <code>mysql</code> , <code>mariadb</code> , or <code>clickhouse</code>
Service name	Oracle only — the service name
Auth Type	<code>sql</code> (username + password appear) or <code>win</code> (Windows/trusted authentication)

Field	Value
User / Password	the monitoring account on the target database (stored encrypted)

3. Click **Test Connection** — a live probe against the target; fix any error before saving.
4. Submit. The server is upserted into `metrics.monitored_servers` and you are redirected to the 7 – *Automation and Workflows* dashboard.
5. **Verify collection:** within a few scheduler cycles the server should show data on the monitoring dashboards. If nothing arrives, check the collection log (Grafana *Operation Log* panels or `log.operation_log`).

Grant the monitoring account read/DMV permissions on the target database (e.g. `VIEW SERVER STATE` on SQL Server).

5.2 Add a Whitelist (Approved) Applicative Login

DBDOME's **App Login Guard** watches program/login combinations: a session on a *watched program* (e.g. SSMS) whose login is **not whitelisted** raises alert `SEC-SQL-ACC-030-RC01` (and can kill the session, subject to the blocker switch). Whitelisted logins (`white = true`) never raise alerts — the suppression is enforced centrally, so **no DBDOME alert of any kind is created for a whitelisted applicative login**.

1. Open `https://<server-ip>:8080/app_login_guard` (or the "Configure App Login Guard" button on the *App Login Guard* Grafana dashboard).
2. Under **Applicative Logins**, enter: - *Login* — the account name; SQL LIKE patterns allowed (`svc_app`, `%app%`) - *Server* — a specific server name, or blank for **all** servers
3. Click **Add**. New entries are whitelisted (`white = true`) by default.
4. Optionally add entries under **Watched Programs** (e.g. `%SSMS%`, `sqlcmd%`) — the guard only enforces when both lists are populated and the Security/critical *blocker* switch is enabled.
5. The Enable/Disable toggle on each row activates/deactivates the entry (`is_active`).

To turn an approved login into a **watched-only** login (alerts fire when it's used from a watched program), clear its `white` flag — this currently has no UI toggle:

```
& "C:\Program Files\PostgreSQL\18\bin\psql.exe" -U dbdome_adm -d dbanalytics -c `
"UPDATE metrics.app_logins SET white = false WHERE applicative_login = 'svc_app';"
```

Related page: `/exclude_logins` — logins listed there (e.g. the DBA team's own accounts) are excluded from self-activity collection and alerting entirely.

5.3 Mail (SMTP) Configuration

1. Open `https://<server-ip>:8080/mailconfiguration` (or the "Email Configuration" button in Grafana).

2. Fill in:

Field	Value
SMTP Server / Port	e.g. <code>smtp.office365.com</code> / <code>587</code>
SMTP User / Password	mailbox credentials (password stored encrypted)
TLS	check for STARTTLS (port 587)
Sender	the From address
Group name / Recipients	a recipient group and its comma-separated address list

3. Click **Send test email** — sends "DBDOME SMTP test" using the values currently on the form *without saving*, and reports success/failure. Fix errors before saving.
4. Submit. Settings are stored in `config.mail_config`, recipients in `config.mail_groups`; alert and scheduled-report mails use them immediately (no service restart needed).

Additional recipient groups can be added later at `/addrecipients`.

6. Updating DBDOME

Updates ship as media containing a `dbdome` folder plus `dbdome_update.exe`.

1. Mount the update ISO / insert the disc (or copy the `dbdome` folder to a drive root).
2. Make sure PostgreSQL is running — the updater checks but does not start it: `powershell Start-Service postgresql-x64-18`
3. Run the updater elevated: `powershell Start-Process -FilePath "E:\dbdome_update.exe" -Verb RunAs -Wait #` or point it at the payload explicitly: `# dbdome_update.exe --source "E:\dbdome"`
4. The updater automatically: stops the DBDOME services → overlays new binaries into `C:\ProgramData\DBDOME\bin` (preserving `DBDOME_SECRET_KEY` and `PG_PASSWORD` in `.env`) → reinstalls and restarts the services → swaps `grafana.db` (your Grafana users are migrated; the old file is kept as `grafana.db.bak.<timestamp>`) → applies all pending SQL migration scripts (tracked in `meta.schema_migrations`, so reruns are safe) → re-encrypts secrets.
5. **Check the end-of-run summary** for any "SQL scripts that did not succeed", and review `C:\ProgramData\DBDOME\update.log`.

7. Post-Install Hardening & Troubleshooting

7.1 Change the default database password

All three PostgreSQL roles are installed with the same initial password. Change it and keep `PG_PASSWORD` in `C:\ProgramData\DBDOME\bin\.env` in sync (the updater/services accept it in plain text and encrypt it to `enc:v1:` on the next update run), then restart the DBDOME services.

7.2 Where to look when something is wrong

Symptom	Check
Installer failed	C:\ProgramData\DBDOME\install.log
Web UI down on 8080	Get-Service DBDOME_web ; restart it; Event Viewer → Application
Dashboards empty	Get-Service DBDOME_scheduler ; Grafana <i>Operation Log</i> ; log.operation_log table
Grafana down on 3000	Get-Service DBDOME_Grafana ; C:\ProgramData\DBDOME\grafana-svc.log
No alert mails	/mailconfiguration → Send test email ; verify recipients in config.mail_groups
Update issues	C:\ProgramData\DBDOME\update.log + the failed-SQL summary

7.3 Service control quick reference (elevated PowerShell)

```
Restart-Service DBDOME_web, DBDOME_scheduler
Restart-Service DBDOME_Grafana
Get-Service DBDOME*, postgresql-x64-18 | Format-Table Name, Status, StartType
```